Equilibrium or Retrieval?

Do LLM Agents Derive or Recall Game-Theoretic Equilibria, and What That Means for Mechanism Design

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Working paper, October 11, 2026

Abstract

Large language models (LLMs) play textbook games well. A model can reach the textbook answer in two ways, however: by *deriving* it from the rules, or by *recognising* a famous game and recalling its answer. The two are behaviourally identical on the textbook question, and they come apart only when the question changes. We separate them with *isomorphic twins*. Every question is asked in its standard wording and in a disguised wording with identical numbers, rules and information structure but no game vocabulary. A third, pre-registered study adds an *extreme* disguise: no story, coded answers and a neutral answer box. Every answer is paired with a competence probe in the same wording, so a change in play cannot be blamed on misreading. Four models from four providers answer 18,768 questions across a Vickrey auction, a Cournot duopoly, a finitely repeated prisoner's dilemma and a public goods game, under three pre-registrations.

The four models split into three patterns:

- **Hidden but still solved.** When the extreme disguise removes recognition of the Cournot game (Nemotron 3 Ultra: 96%→10%; Kimi K3: 72%→20%), their equilibrium play stays at 99–100%, the pattern our pre-registered key reads as consistent with derivation.
- **Graded retrieval.** gpt-oss-120b's Cournot recognition (96%→9%→1%) and equilibrium play (96%→80%→30%) fall together across the three wordings, with every competence probe passed. Its errors become the collusive quantity.
- **Retrieval and comprehension limits.** Gemini 3.5 Flash Lite plays the Cournot equilibrium in 96% of standard questions but in only 4% of disguised ones, and it cannot read the coded extreme rules at all.

In the prisoner's dilemma, recognition retrieves a cultural script of cooperation rather than the equilibrium: removing the familiar frame raises defection in every model (pooled 46%→83%). Fitted to the data, logit QRE fails our pre-registered calibration test for all four models, because LLM errors are rare, large and discrete rather than payoff-scaled noise. [TBD: Stage 2B traps.]

We then ask what noisy, quantal-response agents imply for mechanism design. We prove three results:

- a payment multiplier in a second-price auction is irrelevant for logit-QRE bidders but ruinous for bidders who recall "bid your value", which makes it a test of retrieval;
- the revenue-optimal reserve price is U-shaped in bidder precision but gains at most about 1% over Myerson's;
- the cooperation subsidy that achieves any target cooperation rate has a closed form, whose stage-game halfway point is free of parameters across the entire regular-QRE family.

A pre-registered test pits the derived subsidy against a classically guessed one: [TBD: Stage 5 result].

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1 Introduction

LLM agents are beginning to act in settings that game theory was built to analyse: bidding in auctions, setting prices, negotiating, and drawing on shared resources. The rules of such settings are designed, and designing them, which is the business of mechanism design, requires a model of how participants behave. Classical mechanism design assumes rational players who best-respond. Behavioural mechanism design replaces them with an empirically fitted model, most prominently the logit quantal response equilibrium (QRE) [6, 11], in which players choose better actions more often than worse ones, with a precision λ.

Before a fitted model of LLM play can be trusted to design anything, a prior question must be settled: *is the play we measure reasoning about the rules in front of the model, or retrieval of answers to famous problems?* The concern is concrete. The Vickrey auction, the Cournot duopoly, the prisoner's dilemma and the public goods game are among the most written-about games in any corpus a model is trained on. "Bid your true value", "produce $(a - c)/3b$" and "defect in the last round" appear thousands of times. LLM accuracy is known to depend on how familiar a task looks,

not only on what it requires [10]. A mechanism tuned to recalled play may fail precisely when the designer changes the rules, which is the only reason to design a mechanism at all.

Approach. We hold each game fixed and vary only its *wording*:

(i) **Isomorphic twins.** The disguised wording preserves every number, rule and information structure, and removes the game's vocabulary. An extreme disguise (Stage 2C) removes the story too, codes the answer (for example, "your offer is $3x + 5$") and replaces the answer box. Section 3 shows that a purely deriving agent's play is invariant to such relabelling, so any wording effect rejects pure derivation.

(ii) **Competence probes.** A short calculation in the same wording follows every answer. Only answers whose probe is passed are analysed, so a wording effect cannot be a reading failure.

(iii) **Measured recognition.** A validated code records whether the model's reasoning names the game (agreement with blind hand coding $\kappa = 1.00$). Recognition and play are then read jointly, by rules fixed before the data.

(iv) **Pre-registration.** Three registrations on OSF fix the predictions, sample sizes, exclusions and tests before each study [13]. The analysis code was locked by cryptographic fingerprint.

Findings.

- **Derive or recall differs by model.** Nemotron and Kimi solve a Cournot game they no longer recognise. gpt-oss's play tracks recognition across three wordings. Gemini loses the Cournot equilibrium as soon as the game's name goes.
- **Competence predicts robustness.** The pre-registered ranking held exactly: the higher a model's competence-probe pass rate, the smaller its Cournot drop under disguise.
- **Recognition can pull *away* from equilibrium.** In the prisoner's dilemma, the familiar frame triggers cooperation. Removing it raises defection in all four models, by 17–76 points.
- **Quantal response does not fit LLM play.** It fails held-out calibration for every model, which motivates testing directly whether the models respond to incentives as QRE or as classical theory predicts (Stage 5).

Contributions.

1. A design that identifies derived versus recalled strategic play: isomorphic twins, same-wording competence probes, measured recognition, and an extreme disguise, all pre-registered.
2. Evidence that the answer differs by model and by game, including a retrieval signature in Gemini's Cournot play and recognition-driven cooperation in the prisoner's dilemma.
3. Theory for mechanism design with QRE agents: an invariance theorem that turns a payment multiplier into a test of retrieval; the U-shaped revenue-optimal reserve; and a closed-form cooperation subsidy with a parameter-free prediction. These are tested on the same models in a pre-registered study.

2 Related work

LLMs as economic and strategic agents. Horton et al. [8] proposed treating LLMs as simulated economic agents and reproduced classic behavioural-economics findings with them. Akata

et al. [2] placed LLMs in finitely repeated 2×2 games and found strong play in self-interested games, such as the prisoner's dilemma, and weak coordination. Fan et al. [5] tested basic requirements of rationality. Benchmarks such as GTBench [4] measure strategic reasoning across many game types. These studies vary the game; we vary only the wording of a fixed game, which isolates recognition from ability.

Behavioural game theory for LLMs. Jia et al. [9] evaluate 22 LLMs with behavioural models rather than Nash benchmarks. Pechon-Elkins and Chun [14] audit QRE-based measures of LLM strategic reasoning and conclude that fitted parameters are payoff- and model-dependent summaries rather than structural measures of rationality. We adopt that caution explicitly:

- we never compare λ across games, whose payoff scales differ;
- our main comparisons hold payoffs fixed by construction;
- we verify by parameter recovery that our fitting recovers known λ;
- we judge fit by held-out calibration, decided before the data.

Familiarity, memorisation and reasoning. McCoy et al. [10] show that LLM performance tracks how probable a task and its output are under the training distribution. Our twins apply this idea to strategic play, and the competence probes separate unfamiliarity of the *wording* from failure to *understand* it.

Mechanism design and bounded rationality. Our auction results build on the second-price auction [16] and on Myerson's optimal auction and implementability results [12], in the treatment of Roughgarden [15]. QRE was introduced by McKelvey and Palfrey [11] and is developed in Goeree et al. [6]. Our repeated-game discussion uses Benoît and Krishna [3].

3 A framework for derivation and retrieval

Let a *question* be a pair (G, w) of a game G (players, actions, payoffs, information) and a wording w that presents it. Two wordings w, w' are *isomorphic* for G if there is a bijection ϕ between their answer spaces such that, for every answer a, the payoff consequences of a under w equal those of $\phi(a)$ under w'. For example, "offer $= 3x + 5$" is isomorphic to "bid b" under $b = \phi(x) = 3x + 5$.

Definition 1. A model *derives* on G if the distribution of its answers depends on (G, w) only through G and the answer map: $\Pr(a \mid G, w') = \Pr(\phi^{-1}(a) \mid G, w)$ for every pair of isomorphic wordings. It *retrieves* if its answer distribution depends on features of w that are payoff-irrelevant, such as a game's name.

Proposition 1 (invariance)**.** *For a deriving model, every statistic of play that is defined through payoffs (for example the equilibrium-play rate, or λ fitted on payoffs) is identical across isomorphic wordings. Hence a difference in such a statistic across isomorphic twins, among answers whose competence probe is passed, rejects pure derivation.*

Proof. The statistic is a functional of the answer distribution mapped into payoff space. Under derivation, the mapped distributions coincide by definition. □

Remark. The converse fails. Invariance is also produced by a model that recalls an answer and recognises the game in *both* wordings. So equal play across twins is evidence of derivation only when

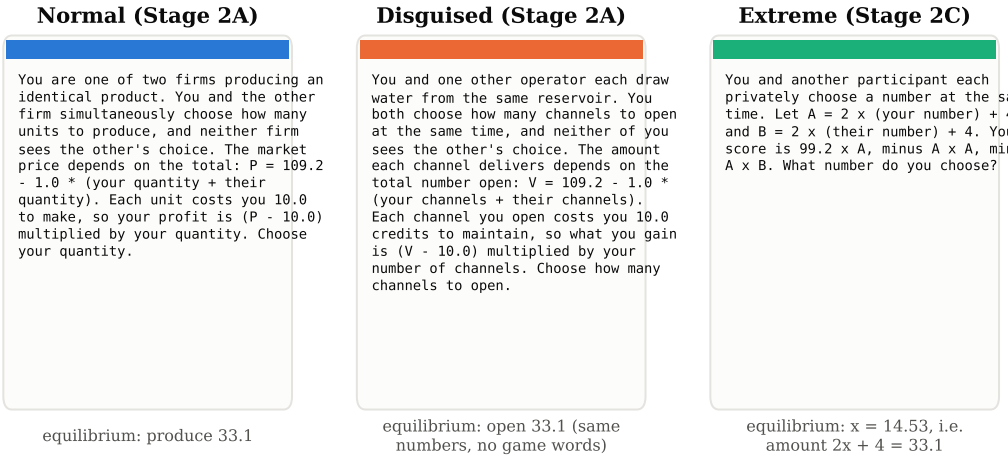


Figure 1: One Cournot question in its three wordings. All three versions have identical numbers and the same equilibrium (in the extreme wording, the coded answer decodes to the same amount). Each answer is followed by a maths check posed in the same wording.

Table 1: The four games. Parameters are drawn per question from seeded samplers.

Game	Parameters	Equilibrium	Disguised as
Vickrey auction	3 bidders, private value $v \sim U[10, 100]$	bid v	research stations requesting slots
Cournot duopoly	$P = a - Q$, $a \in [80, 120]$, $c = 10$	$q = (a - c)/3$	operators opening water channels
Repeated PD	$(T, R, P, S) = (5, 3, 1, 0)$, 10 rounds, round 1	defect	technicians setting panels X/Y
Public goods	n, endowment e, $1 < M < n$	contribute 0	operators routing signal

the disguise actually hides the game. This is why we measure recognition and add an extreme disguise. Our pre-registered interpretation key reads "play unchanged, and recognition drops" as *consistent with* derivation, since recognition is coded from what a model writes, and silent recognition cannot be excluded.

A second route to identification changes the *rules* instead of the wording. If a trap variant has a different equilibrium from the famous game (for example, the winner pays twice the second bid), a retrieving model gives the famous answer where it is wrong. Theorem 1 shows that, for QRE players, this particular change is payoff-irrelevant up to scale, which sharpens the test.

4 Games, wordings and competence probes

Games. Table 1 lists the four games. Each has a unique equilibrium action:

- a weakly dominant strategy in the auction;
- the Cournot–Nash quantity;
- defection by backward induction, since the horizon is common knowledge;

- zero contribution, the dominant strategy, in public goods.

Numbers are drawn per question from seeded samplers, so every question can be reproduced exactly.

Wordings. Each wording has three paraphrases (reworded, or reordered), which never add or drop a fact, so that a disguise effect can be told apart from a wording effect.

- The **disguised** wording moves the game into an unfamiliar story and keeps the numbers, the rules and who knows what.
- The PD's disguised labels `X`/`Y` are counterbalanced, so the cooperative label is randomised.
- The **extreme** wording (Stage 2C) has:
  - no story;
  - coded answers (Vickrey offer $= 3x + 5$ with a private prize; Cournot amount $= 2x + 4$ with the profit written in expanded form);
  - public goods asking how much to *keep*;
  - PD payoffs transformed to $2u - 4$;
  - a neutral answer box.

Each extreme question shares its seed and paraphrase index with a standard question, so every question exists in three wordings with identical numbers (Figure 1).

Competence probes. The probe is a one-step calculation in the question's own wording, for example "you request 50 and the largest other request is 60; what is your net result?". Verdicts:

- AT_EQUILIBRIUM: the probe is passed and the answer is within ± 0.5 of the equilibrium;
- STRATEGIC_GAP: the probe is passed and the equilibrium is missed;
- COMPETENCE_GAP: the probe is failed;
- parse and format failures.

Only the first two count. Server refusals and empty replies are re-asked, never scored.

Models and studies. gpt-oss-120b (via Groq), Nemotron 3 Ultra (NVIDIA), Kimi K3 (Moonshot, served by NVIDIA) and Gemini 3.5 Flash Lite (Google) were each queried with default sampling, September–November 2026. Each model answers:

Study	Registration	Questions per model
Stage 2A (named vs disguised)	osf.io/2uqpj	4 games × 2 wordings × 138 = 1,104
Stage 2B (traps, sweep, endgame)	osf.io/6hz8d	3,036
Stage 2C (extreme disguise)	osf.io/cqnsy	4 games × 138 = 552
Stage 5 (subsidy test)	pre-registration 4	300–400

The cell size of 138 gives 80% power for a 15-point change with Holm correction across four games.

5 Pre-registration and analysis

Registration and locks. Each pre-registration was fingerprinted (SHA-256) in git and registered on OSF before its data existed. Each fixes:

- the predictions;
- the exact sample sizes, as a stopping rule;
- exclusions;
- the tests and multiple-comparison corrections.

The analysis code is fingerprinted too, and it refuses to run on real data unless its fingerprint matches. Ten deviations, all recorded before the affected analysis, are listed in Appendix A. None changes a registered test.

Tests.

- **Stage 2A:** equilibrium play, named vs disguised, by two-proportion z-tests with Holm correction [7]. Public goods, the predicted null, uses an equivalence test (TOST, ± 13 points).
- **Stage 2C:** each extreme question is paired with its twins; exact McNemar tests, Agresti–Min intervals for paired differences [1], and pooling by summing discordant pairs (the Mantel–Haenszel test for matched pairs).

Quantal response fits. In a logit QRE, a player chooses action a with probability $\propto \exp(\lambda\, u(a))$:

- **PD and Cournot:** u is computed at a symmetric fixed point;
- **public goods:** $u(g) = (M/n - 1)g$;
- **Vickrey:** rivals are assumed to bid their equilibrium bids, with values uniform on $[10, 100]$.

We compute the posterior over a log-uniform grid $\lambda \in [10^{-3}, 10^{3}]$. A cell whose evidence keeps rising towards the top of the grid gets only a lower bound, and is flagged as a ceiling. The headline is $\Delta\lambda = \lambda_{\text{named}} - \lambda_{\text{disguised}}$. In a parameter recovery on simulated QRE data, the nominal 95% intervals covered the true λ in 93.75% of 320 fits.

Reasoning codes. A locked codebook codes the stated reasoning. NAMES_GAME requires the reasoning to use a name for the game that the question does not use. Agreement with blind hand coding was $\kappa = 1.00$ on Stage 2A wording and 40/40 on a blind sample in the extreme wording. Error codes for missed answers were validated in two hand-coding rounds:

- COOPERATION_NORM: $\kappa = 0.75$;
- fraction slip: $\kappa = 0.72$;
- one code ("knew and overrode") failed validation ($\kappa = 0.49$) and is not used.

The fraction slip. With a number-only box, a model sometimes derives a fraction (683/60) and writes only part of it (683). A mechanical correction was fixed in pre-registration 3, before any Stage 2C data. It reads the answer as n/d when two conditions hold:

1. the box holds n, or n and d written together, for a fraction n/d in the model's own reasoning;
2. the reasoning ends on that fraction.

It is applied identically to both stages in the Stage 2C comparison, with uncorrected results reported alongside. Pre-registration 1's own analysis is reported uncorrected, as registered.

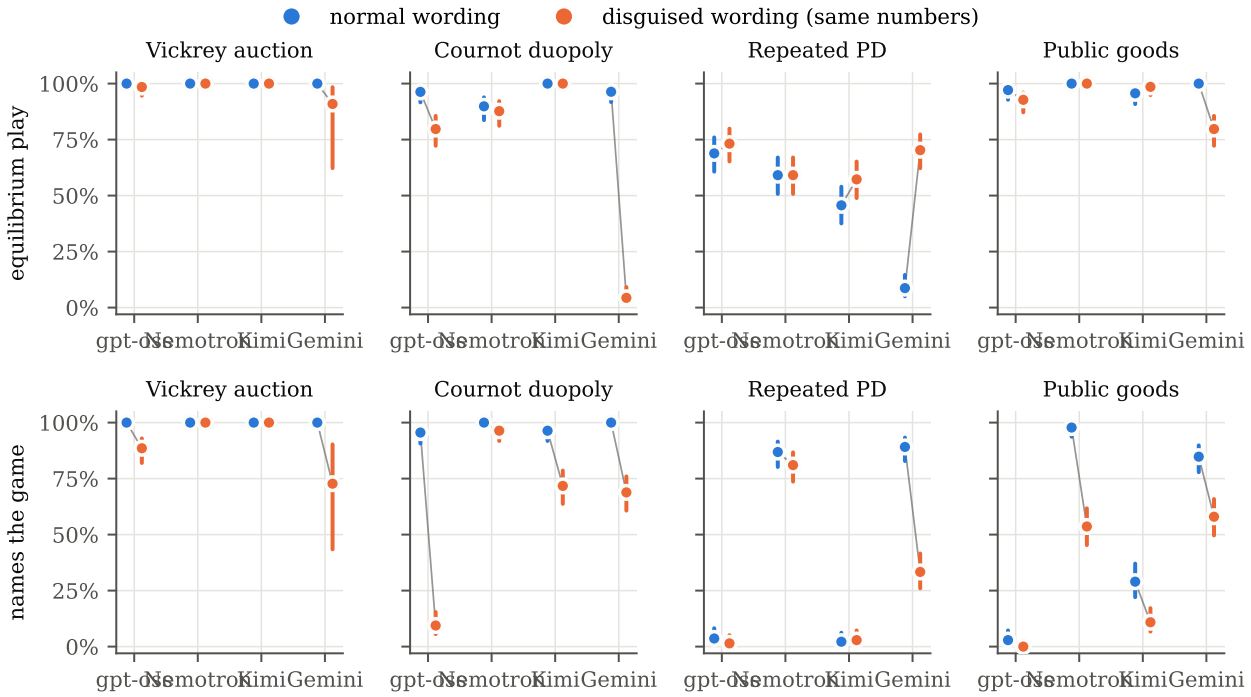


Figure 2: Stage 2A, normal vs disguised wording, with identical numbers, for all four models. Top: equilibrium play among probe-passed answers. Bottom: the share of answers whose reasoning names the game. Bars are 95% Wilson intervals. Gemini's disguised Vickrey cell keeps only 11 answers: the model failed the disguised competence probe on 127 of 138 questions.

6 Results

6.1 Standard twins (Stage 2A)

Table 2 and Figure 2 give the registered test for all four models.

Two models are invariant; two are not.

- **Nemotron and Kimi** play identically in both wordings in every game. Public goods is confirmed equivalent within ±13 points for both (TOST $p < 10^{-30}$ and $p = 2 \times 10^{-6}$). By Proposition 1 this is consistent with derivation. It is not proof, however: both models often name the disguised game (Figure 2, bottom). Nemotron names it in 96% of disguised Cournot answers and 100% of disguised auctions, so for these models the standard disguise does not hide the game, which motivates Stage 2C.
- **gpt-oss-120b** loses 17 points of Cournot equilibrium play under disguise (96% → 80%, Holm $p = 10^{-4}$), with all other games unchanged and public goods confirmed equivalent (TOST $p = 8 \times 10^{-4}$). This falls inside the range pre-registration 1 committed to for this model (−15 to −25). The fits agree: λ falls from 0.173 to 0.027 ($\Delta\lambda = +0.146$, $P(\Delta\lambda > 0) = 1.00$).
- **Gemini** shows the largest effects, described next.

Gemini shows a retrieval signature in Cournot. Gemini plays the Cournot equilibrium in 96% of named questions and in 4% of disguised ones, a 92-point drop, yet passes the disguised

Table 2: Stage 2A main test (pre-registration 1): equilibrium play among probe-passed answers, named vs disguised. Holm correction across the four games within each model.

Model	Game	Named	Disguised	Change	Holm p
gpt-oss-120b	Vickrey	100% (138/138)	98% (129/131)	−2	0.30
	Cournot	96% (130/135)	**80% (110/138)**	**−17**	1×10^{-4}
	Repeated PD	69% (95/138)	73% (101/138)	+4	0.43
	Public goods	97% (134/138)	93% (128/138)	−4	0.30
Nemotron	Vickrey	100% (138/138)	100% (138/138)	+0	1
	Cournot	90% (124/138)	88% (121/138)	−2	1
	Repeated PD	59% (81/137)	59% (81/137)	+0	1
	Public goods	100% (136/136)	100% (138/138)	+0	1
Kimi	Vickrey	100% (138/138)	100% (138/138)	+0	1
	Cournot	100% (138/138)	100% (138/138)	+0	1
	Repeated PD	46% (63/138)	57% (79/138)	+12	0.22
	Public goods	96% (132/138)	99% (136/138)	+3	0.45
Gemini	Vickrey	100% (138/138)	91% (10/11)	−9	4×10^{-4}
	Cournot	96% (132/137)	**4% (6/138)**	**−92**	6×10^{-52}
	Repeated PD	9% (12/138)	**70% (97/138)**	**+62**	4×10^{-25}
	Public goods	100% (138/138)	80% (110/138)	−20	5×10^{-8}

competence probe in all 138 questions. It can compute profits in the disguised wording; it does not find the equilibrium there. Its disguised misses are mostly coded as a claimed but incorrect Nash calculation (44) or the collusive (joint-monopoly) quantity (31). The fits agree: λ falls from 0.649 $[0.50, 0.82]$ to 0.0047 $[0.0036, 0.0060]$.

The auction drop is a reading failure, not a strategic one. Gemini's disguised Vickrey cell keeps only 11 answers, because it failed the disguised probe in 127 of 138 questions. The design classifies this correctly, as a competence gap, so it is not evidence about strategy. Without the probes it would read as a 92-point strategic collapse.

In the prisoner's dilemma, recognition pulls away from equilibrium.

- **Gemini:** with the game named, it cooperates in round 1 of a known 10-round game in 91% of questions; disguised, it defects in 70%. Every one of its 126 named-wording cooperative answers gives a cooperation-norm reason.
- **Kimi and gpt-oss** move the same way, by +12 and +4 points (not significant).

Recognition retrieves a *cultural* script of cooperation in repeated dilemmas, not the backward-induction solution. Pre-registration 1 predicted this direction and named Gemini as the model with the largest swing; both predictions held.

Public goods. Gemini contributes more when the game is disguised (100%→80%; equivalence not shown), and most of its disguised misses argue for the group optimum (20 of 28).

Pre-registration 1, scored.

- **Cournot drop in every model:** the direction holds for gpt-oss and Gemini, but not for Nemotron (−2) or Kimi (0).

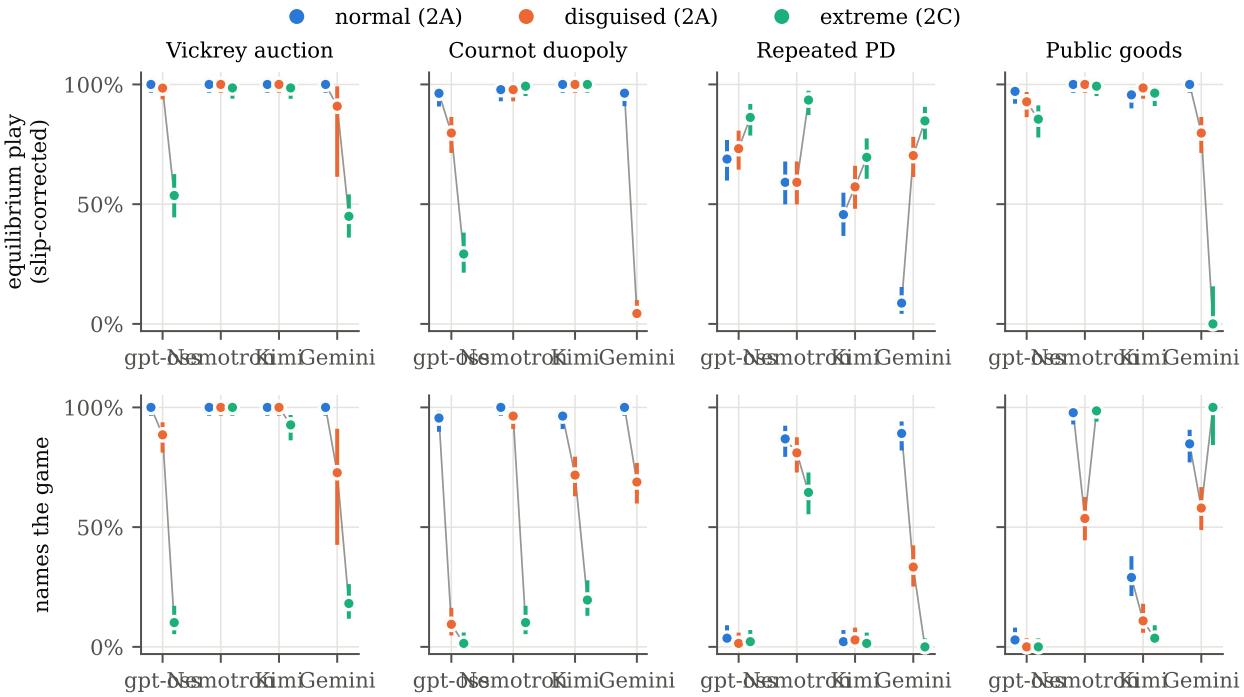


Figure 3: The same questions in three wordings, for all four models. Top: equilibrium play (fraction-slip corrected, both stages alike). Bottom: recognition. Gemini's extreme Cournot cell is empty: it failed the extreme competence probe in all 138 questions.

- **PD rise under disguise:** significant only for Gemini; Gemini having the largest swing was predicted correctly.
- **Public goods flat:** met for three of four models (not Gemini).
- **Ranking prediction:** the model with the highest competence-probe pass rate shows the smallest Cournot drop. This held *exactly*: Kimi 100% and 0 points, Nemotron 99.6% and −2, gpt-oss 99.4% and −17, Gemini 88.4% and −92.

6.2 The extreme disguise (Stage 2C)

The extreme disguise removes the story, codes the answer and neutralises the answer box. Table 3 and Figure 3 give the registered comparison with the same questions in standard wording. Pooled over models, play changes in every game. The pooled figures conceal three distinct patterns, however, and the pre-registered interpretation key reads each one differently.

Pattern 1: hidden but still solved (Nemotron, Kimi). The extreme wording largely removes recognition of the Cournot game:

- Nemotron: 96%→10% ($p = 3 \times 10^{-36}$);
- Kimi: 72%→20% ($p = 5 \times 10^{-16}$).

Equilibrium play nevertheless stays at 99–100%. The key reads this as *consistent with derivation*: both models solve a game they no longer name.

Table 3: Stage 2C (pre-registration 3): equilibrium play, normal (2A) vs extreme (2C), on paired questions with identical numbers. Exact McNemar test, Holm across games within each model; the pooled rows sum discordant pairs over models (Mantel–Haenszel for matched pairs).

Model	Game	Pairs	Normal	Extreme	Normal − extreme [95% CI]	Holm p	Names game[†]
gpt-oss	Vickrey	138	100%	54%	+46 [+37, +54]	3×10^{-19}	89%→10%
	Cournot	134	96%	30%	+66 [+57, +74]	4×10^{-24}	9%→1%
	Repeated PD	138	69%	86%	−17 [−28, −7]	3×10^{-3}	1%→2%
	Public goods	138	97%	86%	+12 [+5, +18]	3×10^{-3}	0%→0%
Nemotron	Vickrey	138	100%	99%	+1 [−1, +4]	1	100%→100%
	Cournot	138	98%	99%	−1 [−5, +2]	1	96%→**10%**
	Repeated PD	137	59%	93%	−34 [−43, −25]	3×10^{-10}	81%→65%
	Public goods	136	100%	99%	+1 [−1, +3]	1	54%→99%
Kimi	Vickrey	137	100%	99%	+1 [−1, +4]	1	100%→93%
	Cournot	138	100%	100%	+0 [−1, +1]	1	72%→**20%**
	Repeated PD	138	46%	70%	−24 [−34, −13]	3×10^{-4}	3%→1%
	Public goods	138	96%	96%	−1 [−5, +4]	1	11%→4%
Gemini	Vickrey	138	100%	45%	+55 [+46, +63]	5×10^{-23}	73%→27%[‡]
	Cournot	0	no extreme answer passed the probe (0/138)				
	Repeated PD	138	9%	85%	−76 [−83, −67]	4×10^{-27}	33%→0%
	Public goods	22	100%	0%	+100 [+78, +100]	5×10^{-7}	50%→100%[‡]
All models	Vickrey	551	100%	74%	+26 [+22, +30]	4×10^{-43}	96%→67%
	Cournot	410	98%	77%	+21 [+17, +25]	2×10^{-21}	59%→10%
	Repeated PD	551	46%	83%	−38 [−43, −33]	1×10^{-36}	30%→17%
	Public goods	434	98%	89%	+9 [+5, +12]	3×10^{-7}	23%→37%

[†] disguised (2A) → extreme (2C), on the same paired questions. [‡] small n (11 and 22 pairs): most of Gemini's disguised or extreme answers failed the probe.

Pattern 2: graded retrieval (gpt-oss). gpt-oss passes the competence probe in the extreme wording, yet its play falls sharply: Vickrey 100% → 54%, Cournot 96% → 30%. In Cournot, recognition and play fall together across the three wordings:

	Normal	Disguised	Extreme
Names the game	96%	9%	1%
Equilibrium play	96%	80%	30%

Within the extreme auction, gpt-oss plays the equilibrium in 14 of 14 answers that name the game, against 60 of 124 that do not (Fisher $p = 10^{-4}$, exploratory). Its errors are systematic, not noisy:

- 67 of its 97 Cournot misses are the *collusive* quantity, the joint optimum: without the textbook cue, it treats the game as cooperative;
- its auction misses are corner bids (0, ±1000).

No miss is a conversion slip (a strategy left in the wrong unit). The key reads this as *recall*: the play depends on recognising the game, and the competence checks rule out a reading failure.

Pattern 3: the coded rules defeat comprehension (Gemini). Gemini failed the extreme competence probe in all 138 Cournot questions and in 116 of 138 public-goods questions. These cells say nothing about strategy. The key reads them as a comprehension cost, which the design was built to detect.

The prisoner's dilemma, in every model. Removing the familiar frame raises defection for all four models:

- Nemotron +34 points;
- Kimi +24;
- gpt-oss +17;
- Gemini +76;
- pooled: 46% → 83%, $p = 10^{-36}$.

Kimi almost never *names* the game in any wording (1–3%), so the cooperation pull is carried by the frame itself (cooperate/defect labels and a scored payoff table), not by the label "prisoner's dilemma". The only pre-registered pooled prediction that held exactly was this one (extreme defection 80–90%, above the normal-wording rate).

The fraction slip was not a detail. In the extreme wording, coded answers are usually awkward fractions, so the slip becomes common. For Nemotron, 50% of extreme Vickrey answers and 55% of Cournot answers were written as a numerator or a concatenation. Without the correction fixed in advance, Nemotron would show spurious drops of 51 and 46 points (both $p < 10^{-18}$); with it, the drops vanish.

Pre-registration 3, scored. The author's predictions were mostly wrong, and wrong in an informative direction:

- **Pooled recognition:** predicted ranges were not met for any game.
- **Pooled play:** the Vickrey and Cournot drops were real, but of different sizes from the predictions.
- **Per-model Cournot drops:** the author expected Nemotron to lose Cournot (≥ 35 points) and gpt-oss and Kimi to be stable. Nemotron and Kimi were stable (-1 and 0), while gpt-oss lost 66 points; Gemini could not be tested.

The model the author believed was recalling is the one that derives, and vice versa.

6.3 Does quantal response describe the models? The decision gate

Pre-registration 1's gate asks whether the QRE fits are good enough to design mechanisms on. Its criteria were fixed by the author before three of the four models' fits existed:

- fit λ on two of the three normal-wording paraphrases;
- predict the held-out paraphrase's equilibrium-play rate;
- call a game calibrated if the prediction is within ± 10 points in at least two of three rotations;
- call a model's fits good if at least three of its four games are calibrated.

By these criteria, fits are poor for all four models: 0, 1, 1 and 1 of 4 games calibrated. The Vickrey cells fail for an artifactual reason: at the top of the registered λ grid, QRE predicts only 82–90% equilibrium play, while every model plays it 100%. The verdict does not depend on this, though: counting Vickrey as calibrated, no model reaches three games.

The substantive reason is visible in the data. LLM departures from equilibrium are *rare, large and discrete*:

- fraction slips;

- collusive quantities;
- corner bids;
- a norm-driven preference for cooperation.

They are not the frequent, small, payoff-scaled deviations that logit QRE describes. A handful of far-off answers drags the fitted λ down, and the fitted model then badly under-predicts the exact-equilibrium rate (for example, 27–51% predicted against 91–100% observed for gpt-oss in Cournot). Two models also fail a hard check in the prisoner's dilemma: they cooperate more than 50% of the time with no subsidy, which no stage-game QRE can produce.

The gate therefore selects option 3: the behavioural model does not describe the agents, and the mechanism stage studies why. Stage 5 (Section 8) does this directly, by asking whether the agents respond to incentives as QRE or as classical theory predicts.

6.4 Stage 2B: traps, number sweeps and the endgame

[TBD: Block B (traps): the correct share and the recall signature (the textbook answer where it is wrong), named vs disguised, CMH across models. Block C (sweep): the wording×size interaction. Block E (endgame): defection at rounds 5, 9 and 10. Complete for three models; gpt-oss-120b's blocks are still in collection.]

7 Mechanism design for noisy agents

We now treat each player as a logit-QRE player with precision λ and ask how rules should change. All results are verified numerically: the code accompanies the paper, and every claim has a test.

7.1 Second-price auctions with a reserve and a payment multiplier

Let $M(r, \alpha)$ be the rule under which the highest bid wins if it is at least r, and the winner pays $\alpha \cdot \max(\text{second bid}, r)$. The Vickrey auction is $M(0, 1)$, and under $M(0, \alpha)$ the dominant strategy is $b = v/\alpha$. In a symmetric logit QRE, each bidder's bid density is $\propto \exp(\lambda\, U(b \mid v))$, where U is the expected payoff against the other bidders' QRE bids.

Theorem 1 (α is irrelevant for QRE bidders)**.** *If $(B_1, \ldots, B_n)$ is a QRE of $M(r, 1)$ on bid space $[0, \bar{B}]$, then $(B_1/\alpha, \ldots, B_n/\alpha)$ is a QRE of $M(r/\alpha, \alpha)$ on $[0, \bar{B}/\alpha]$, with the same winner, the same revenue and the same payoffs.*

Proof. Against a highest rival bid H, $U_\alpha(b \mid v) = \mathbb{E}[(v - \alpha \max(H, r/\alpha))\mathbf{1}\{H < b,\ b \geq r/\alpha\}]$. Suppose rivals play $H = H_1/\alpha$, and substitute $b = \beta/\alpha$. Then $U_\alpha(\beta/\alpha \mid v) = \mathbb{E}[(v - \max(H_1, r))\mathbf{1}\{H_1 < \beta,\ \beta \geq r\}] = U_1(\beta \mid v)$. So the M_α bid has density $\propto \exp(\lambda U_1(\alpha b \mid v))$. By a change of variables, αB has density $\propto \exp(\lambda U_1(x \mid v))$, which is the M_1 QRE density, and the normalising constants agree because both integrate to one. The scaled profile is therefore self-consistent, which makes it a QRE. Dividing every bid by α preserves their order, and revenue is $\alpha \cdot B_{(2)}/\alpha = B_{(2)}$. □

The invariance needs the bid space to scale with α. Our solver lays the bid grid out in value units and confirms the identity to 10^{-9}.

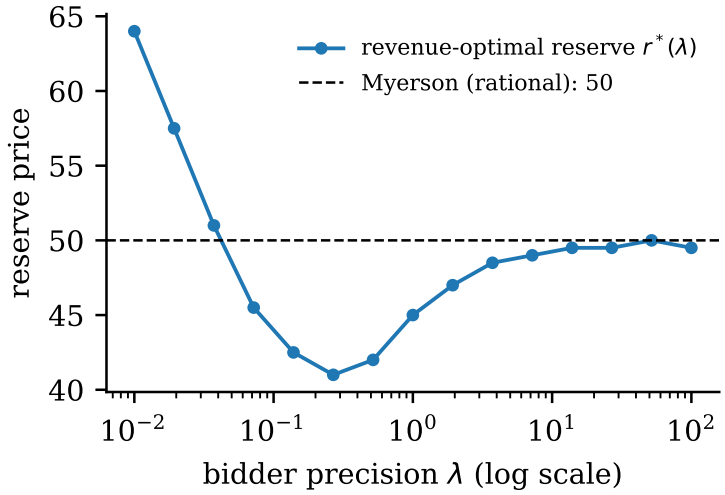


Figure 4: The revenue-optimal reserve for three logit-QRE bidders (values uniform on $[10, 100]$). It is U-shaped in precision and returns to Myerson's 50 as $\lambda \to \infty$.

A test of retrieval. For a deriving QRE bidder, $\alpha \cdot b$ has the same distribution at every α. A bidder that recalls "bid your value" bids v regardless. Under $\alpha = 2$, three such bidders leave the winner with $\mathbb{E}[v_{(1)}] - 2\,\mathbb{E}[v_{(2)}] = 77.5 - 110 = -32.5$. Any dependence of $\alpha \cdot b$ on α therefore cannot be QRE noise; it is the fingerprint of recall. Stage 2B's "winner pays twice" trap implements exactly this test: [TBD: trap result].

Theorem 2 (a reserve never raises efficiency)**.** *For any λ, expected efficiency (the winner's value, or 0 without a sale) cannot be increased by $r > 0$.*

Proof. For fixed bids, r never changes who has the highest bid; it can only cancel the sale, replacing a value of at least 10 by 0. Under QRE the bids also respond to r. That strategic effect is verified numerically: efficiency falls in r at every λ tested. □

The originally planned problem, maximising efficiency over (r, α), therefore has a trivial answer. For *revenue*, with values uniform on $[10, 100]$, rational bidders have $\varphi(v) = 2v - 100$, so Myerson's reserve is $r^* = 50$. For QRE bidders we solve the full symmetric fixed point, in which each bidder responds to the others' noisy bids, and optimise r (Figure 4).

The optimal reserve $r^*(\lambda)$ is U-shaped:

- **Moderate noise** ($\lambda \approx 0.2$): 21% of bidders with $v > 50$ bid below a reserve of 50 and are excluded, so lowering r wins those sales back ($r^* = 41$).
- **Extreme noise** ($\lambda \approx 0.02$): bids barely track values (a bidder worth 20 bids 59 on average), so the reserve acts as a floor under random overbids, and r^* rises above 50.

The revenue gain over Myerson's reserve never exceeds about 1%. Moreover, all three models analysed so far bid at the ceiling of precision in the standard auction. For these agents the auction leaves little to design, so we turn to the game where they are genuinely noisy.

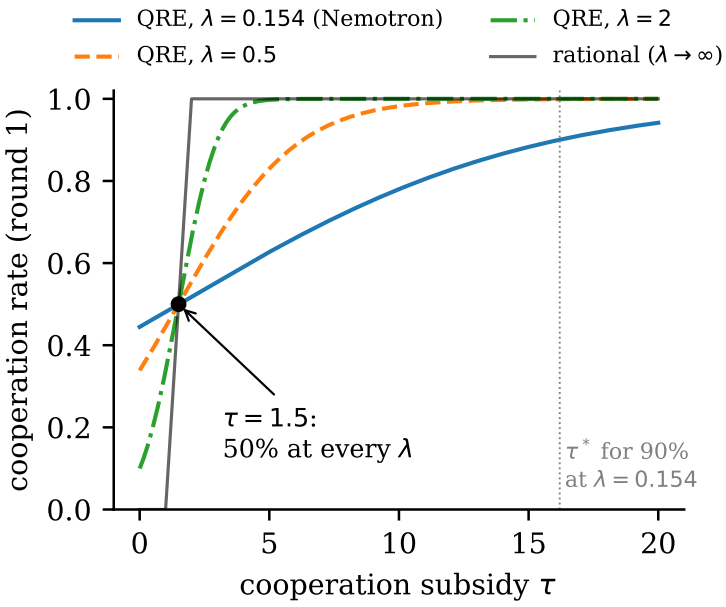


Figure 5: Cooperation under a subsidy τ for logit-QRE players. Every precision passes through 50% at $\tau = 1.5$ (Corollary 1), as does the rational mixed equilibrium. The dotted line is Theorem 3's subsidy for 90% cooperation at Nemotron's fitted $\lambda = 0.154$.

7.2 A cooperation subsidy for the prisoner's dilemma

Under $S(\tau)$, every player who cooperates is paid an extra τ per round. Against an opponent who cooperates with probability p, defecting pays more by $\Delta U(p) = p(T-R)+(1-p)(P-S)$. Under $S(\tau)$ the advantage becomes $\Delta U(p) - \tau$, and the symmetric logit QRE satisfies $p = 1/(1 + e^{\lambda(\Delta U(p)-\tau)})$.

Theorem 3 (closed-form subsidy)**.** *The subsidy at which the symmetric logit-QRE cooperation rate equals p^* is*

$$\tau(p^*, \lambda) = \Delta U(p^*) + \frac{\operatorname{logit}(p^*)}{\lambda}.$$

Since $d\tau/dp = (T - R) - (P - S) + 1/(\lambda p(1 - p)) > 0$, each τ gives exactly one QRE.

Proof. Rearranging, $(1 - p)/p = e^{\lambda(\Delta U(p)-\tau)}$, so $-\operatorname{logit}(p) = \lambda(\Delta U(p) - \tau)$. Solve for τ. □

Corollary 1 (a parameter-free halfway point)**.** *In the stage game, at $\tau_{1/2} = \Delta U(\frac{1}{2}) = \frac{1}{2}[(T - R) + (P - S)]$ (that is, 1.5 for our payoffs), every regular QRE population cooperates with probability exactly $\frac{1}{2}$, whatever its precision and whatever its symmetric error distribution.*

Proof. At $p = \frac{1}{2}$ and $\tau = \tau_{1/2}$ the player is indifferent, so a regular quantal responder randomises evenly. Uniqueness holds because $\Delta U(p) - \tau$ rises in p, while the response falls in it. □

Rational players and the repeated game.

- **Rational stage game:** players defect for $\tau < P - S = 1$ and cooperate for $\tau > T - R = 2$. In between, the subsidised stage game is chicken, whose symmetric equilibrium mixes with $p = \tau - 1$, so it too passes through $\frac{1}{2}$ at $\tau = 1.5$.

- **Repeated game:** chicken has two pure equilibria that pay each player differently, so in the 10-round game, round-1 cooperation can be subgame-perfect: play (C, C), and punish a deviator in the last round with the equilibrium worst for them [3]. At $\tau = 1.5$, deviating gains 0.5, while the punishment costs at least 1.5. So cooperation above 50% at $\tau = 1.5$ is consistent with rational repeated-game play as well as with a cooperation norm, and the reasoning codes separate the two.

Noisy players are cheap to nudge towards partial cooperation and expensive to push towards certainty (Figure 5). At Nemotron's $\lambda = 0.154$:

- 50% cooperation costs $\tau = 1.5$;
- the classical choice of $\tau = 2.5$, just above dominance, yields only about 54%;
- 90% requires $\tau \approx 16.2$.

8 A confirmatory test of the derived mechanism

Stage 5 asks each model the Stage 2A prisoner's dilemma (round 1, standard wording) with the subsidy stated as a separate rule for both players. There are 100 questions per level, balanced across the paraphrases, at four levels:

- $\tau = 0$: the baseline;
- $\tau = 1.5$: Corollary 1;
- $\tau = 2.5$: the classical guess;
- $\tau^* = 1.9 + \ln 9/\hat{\lambda}_m$: Theorem 3's subsidy for 90%, at model m's fitted precision; the derived mechanism.

Because $\tau^* > 2.5$ whenever $\lambda < 3.66$, "the derived subsidy yields more cooperation" would test nothing. The registered comparison is between the theories' *predictions*:

- at $\tau = 2.5$, is QRE's prediction closer to the observed rate than classical theory's 100%?
- at τ^*, is the observed rate within 90 ± 15%?

Decision-gate criteria (held-out calibration) and an interpretation key for every pattern were fixed before the remaining fits existed. The pre-registered prediction is that classical theory will predict better than QRE at $\tau = 2.5$ for Nemotron and Kimi, because their low prisoner's dilemma λ reflects a cooperation norm, not insensitivity to payoffs.

[TBD: Results: cooperation by level and model; QRE vs classical at $\tau = 2.5$; the derived mechanism at τ^*; the halfway-point test.]

9 Discussion

"Derive or recall" has no single answer. The same question receives different answers across models and games:

- Nemotron's and Kimi's Cournot play survives the removal of every cue that could trigger recall;
- gpt-oss's play degrades step by step as the cues are removed, and its errors become the cooperative (collusive) answer;
- Gemini's Cournot play does not survive the removal of the game's name;

- for games whose rules are their fingerprint (the second-price auction, public goods), no wording hid the game from the strongest models, so derivation and recall remain observationally equivalent there.

The last point is a limit of the method, not a failure of it: recognising a game's structure is itself a form of reasoning. More striking is that the author's own pre-registered expectations about *which* models recall were reversed. Intuition about which LLMs "really reason" is unreliable, and that is a reason to measure it.

Competence buys robustness. The pre-registered ranking held exactly: models that pass more competence probes lose less when the game is disguised. A model that can reliably execute the rules has less need to lean on recognising them. This suggests a practical screen for agents that will face unfamiliar rules: probe pass rates under paraphrase are a cheap proxy for robustness to rule changes.

Recognition can pull away from equilibrium. The usual worry is that recall makes models look more rational than they are. In the prisoner's dilemma we find the reverse. Recognising the game lowers equilibrium play: Gemini by 62 points under the standard disguise, and every model by 17–76 points relative to the extreme wording. The retrieved content is the folk lesson of repeated dilemmas (trust, reciprocity, establishing cooperation), not the backward-induction solution. For mechanism design this matters directly: an agent's behaviour depends on which famous game the designer's rules resemble.

Probes change conclusions. Without competence probes, Gemini's disguised auction would read as a 92-point strategic collapse. With them, it is correctly a reading failure. Without a registered slip rule, Nemotron's extreme-wording Cournot and auction results would read as 46- and 51-point drops. Both safeguards altered headline numbers, which is the strongest argument for building them in before the data.

Implications for designing mechanisms for AI agents. Theorem 1 gives a cheap diagnostic: rescale the payment rule and check whether bids rescale. Theorem 3 shows that noise makes high compliance expensive, so subsidies should be set from measured responsiveness rather than from dominance thresholds. And the decision gate warns against borrowing the behavioural model wholesale. Logit QRE fails held-out calibration for every model, because LLM errors are rare, large and discrete: slips, collusive answers and norms. A fitted λ then encodes a norm or a slip rate rather than noise, and a mechanism tuned to it may systematically over- or under-pay. Stage 5 tests this directly.

10 Limitations

- **Disguises can fail.** The standard disguise did not hide the games from two of three models, and no wording hid the auction or public goods. Recognition is coded from written reasoning, so silent recognition cannot be excluded.
- **A leaky answer box.** In Stage 2A's disguised wording the answer field still carried the named game's field names ("bid", "quantity", "contribution"). This was found after collection, is recorded as deviation 8, and was removed in Stage 2C.

- **Timing.** Stage 2C was collected weeks after Stage 2A, so a provider updating a model in between would appear as a difference.
- **The behaviour model.** QRE treats every departure from equilibrium as payoff-scaled noise. Our models' departures are often slips or norms instead, so λ is a summary, not a structural parameter [14], and it is never compared across games.
- **The λ grid.** The registered grid tops out at $\lambda = 1000$, where QRE cannot reproduce 100% equilibrium play in the auction. This affects the calibration of those cells, but not the gate's verdict.
- **The stage-game QRE in the prisoner's dilemma.** It ignores continuation play, which is harmless when the stage equilibrium is unique but not at $\tau = 1.5$.
- **Scope.** Four models, four games and free API tiers with default sampling. Cross-model patterns are rankings, not estimates of population effects.

11 Conclusion

Whether an LLM agent derives or recalls a game's equilibrium is an empirical property of the model and the game, and it can be measured. Isomorphic twins with same-wording competence probes, measured recognition and an extreme disguise separate the two wherever the game can be hidden at all. On that evidence:

- two frontier models solve a Cournot game they no longer recognise;
- a third loses the equilibrium step by step as the cues are removed;
- a fourth loses it as soon as the name goes;
- in the prisoner's dilemma, recognising the game pulls every model away from equilibrium, towards cooperation;
- the standard behavioural model, logit QRE, does not describe any of them.

For mechanism design these are not curiosities. They determine whether a rule tuned to observed play will survive a change in the rules. [TBD: one sentence on Stage 5 once available.]

Acknowledgements

Software development, data collection and drafting were assisted by Claude (Anthropic).

Data and code availability

Code, prompts, seeds, raw answers, analysis outputs and the locked analysis plans are available at [TBD: public repository URL]. Pre-registrations: osf.io/2uqpj, osf.io/6hz8d, osf.io/cqnsy.

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A Deviations from the pre-registrations

1. **PD scoring error in practice runs.** Named answers were scored against the disguised labels; fixed before Stage 2.
2. **Gemini 3.8 Flash replaced** by Gemini 3.5 Flash Lite, because of a 20-request daily cap.
3. **Empty replies** (bursts of "!!!!") are re-asked rather than scored as unreadable.
4. **gpt-oss's daily token cap** slows its collection; this does not affect the design.
5. **A 10-minute NVIDIA access outage.** Its 4,454 refused requests were relabelled as infrastructure failures and re-asked.
6. **The fraction slip.** Recorded as an exploratory observation, and corrected only in the pre-registration-3 comparison.
7. **Block D (reasoning effort)** was dropped before any of its data existed, so that all models answer the same questions.
8. **The disguised answer boxes** carried the named game's field names (a limitation).
9. **Ten questions re-asked** that should have been scored as the model's own (1 Kimi Stage 2A; 2 Nemotron Stage 2A; 7 Nemotron Stage 2B traps). The re-asked answers stand.
10. **The pilot answer form** sent explanatory text that leaked words; fixed before Stage 2C.

B Stage 2A precision fits

Table 4: Posterior λ (median and 95% interval), named vs disguised. "$\geq$" marks a cell at the ceiling of the grid, where only a lower bound is identified. λ is on each game's payoff scale and is not comparable across games.

Model	Game	λ named	λ disguised	$\Delta\lambda$
gpt-oss	Vickrey	≥ 944	1.12 [0.87, 1.45]	$\geq +943$
	Cournot	0.173 [0.13, 0.22]	0.027 [0.022, 0.035]	$+0.146$, $P(>0) = 1.00$
	PD	0.579 [0.27, 0.92]	0.772 [0.45, 1.12]	-0.19, $P(>0) = 0.20$
	Public goods	3.76 [3.07, 4.34]	1.19 [1.00, 1.37]	$+2.53$, $P(>0) = 1.00$
Nemotron	Vickrey	≥ 944	≥ 944	both at ceiling
	Cournot	0.0012	0.0010	$+0.0001$, $P(>0) = 0.82$
	PD	0.154 [0.002, 0.47]	0.188 [0.002, 0.50]	-0.02, $P(>0) = 0.44$
	Public goods	≥ 21.8	≥ 21.8	both at ceiling
Kimi	Vickrey	≥ 944	≥ 944	both at ceiling
	Cournot	≥ 299	19.4 [14.1, 26.6]	$\geq +272$
	PD	0.0075 [0.001, 0.11]	0.079 [0.001, 0.39]	-0.06, $P(>0) = 0.21$
	Public goods	2.51 [2.11, 2.90]	3.98 [3.35, 4.73]	-1.51, $P(>0) = 0.00$
Gemini	Vickrey	≥ 944	3.98 [1.5, 7.7] ($n = 11$)	$\geq +936$
	Cournot	0.649 [0.50, 0.82]	0.0047 [0.004, 0.006]	$+0.645$, $P(>0) = 1.00$
	PD	0.0032 [0.001, 0.023]	0.631 [0.33, 0.97]	-0.629, $P(>0) = 0.00$
	Public goods	≥ 21.8	0.434 [0.36, 0.53]	$\geq +21.2$

Nemotron's Stage 2A Cournot fit is dominated by uncorrected fraction slips (answers such as 683 for 683/60), which are scored as they were written, as pre-registration 1 requires. With the slip correction fixed in pre-registration 3, its Cournot λ is 0.0082 [0.0063, 0.0103] (Table 3). Kimi's

Cournot difference arises entirely within the ± 0.5 tolerance: both wordings are at 100% equilibrium play.